

S&P Global & Crisil Campus Hackathon 2026

# AI/NLP Risk Signal Platform

From news and social text to sentiment, event and impact signals, driving a tactical index rebalancer and an event-driven stress tester

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47

signals

2

modules

8

event types

# Problem and approach

Analysts read far more headlines and posts than they can handle, yet trading and capital decisions need clear numbers they can compare. I wanted to turn raw text into those numbers quickly.

**My approach: build one NLP engine whose output I can explain, then feed it into two different tools: a fast index rebalancer and a slower portfolio stress tester.**

1

## Ingest

News + social, cleaned, de-duplicated, credibility-weighted

2

## Analyse

Sentiment -1..1, 8 event types, impact 1..10

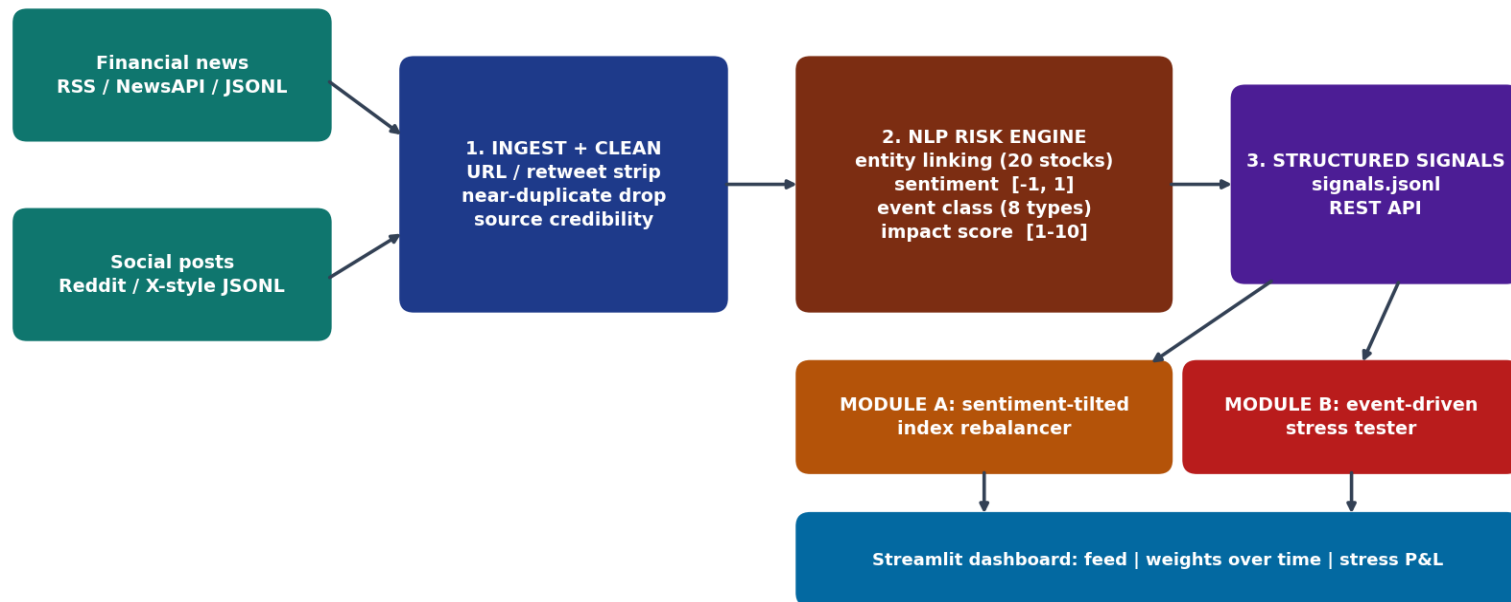
3

## Act

Rebalance weights; stress the bank book

# System design and data flow

## System architecture & data flow



# Implementation



My Streamlit dashboard, running on the sample data. The sidebar controls the rebalancer.

47

signals

7

high impact

2

sources

## NLP engine

Word lexicon with negation, regex-based event types, and impact =  $0.55 \times \text{event prior} + 3 \times |\text{sentiment}| + 2 \times \text{intensity} + 1.5 \times \text{source credibility}$ .

## Module A: Rebalancer

Each stock keeps a sentiment score that fades with a 90 min half-life. Weights tilt up or down, kept between 2% and 12%.

## Module B: Stress test

If impact is above 7 and sentiment is negative, shocks (equity, rates, spreads, FX, defaults) are applied to a 36-position bank book.

# Results



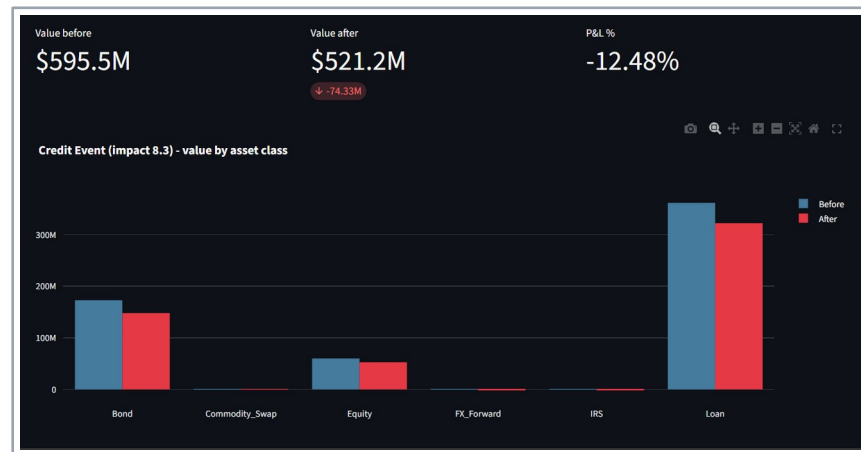
Module A: weights shift through the day as news arrives. GOOGL gains 4.7 points, AAPL loses 2.2.

47

signals from 48 docs

7

stress triggers



Module B: a Credit Event at impact 8.3 takes the book from \$595.5M to \$521.2M, a 12.48% loss.

2.2%

avg turnover per step

5.6 bps

total trading cost

# Domain impact

1

## Speed

A headline becomes a risk number in seconds instead of hours of reading

2

## One signal, two horizons

The same feed supports quick index tilts and longer-term stress planning

3

## Explainable

Each score can be traced to specific words and a written formula, so a reviewer can check it

4

## Extensible

New data sources or a FinBERT model can be added without redoing the rest

# Limitations and next steps

## Limitations

- Lexicon and regex miss sarcasm and context
- Demo corpus is synthetic and small
- Rebalancer not backtested on real prices
- Stress shocks are illustrative, not calibrated

## Next steps

- Fine-tune FinBERT on labelled finance text
- Backtest vs equal-weight with market data
- Streaming ingestion (Kafka) and alerts
- Factor-based and reverse stress tests